

CPSC 417: Distributed Systems and Network Applications

Winter Term 2, 2027

**We acknowledge the use of OpenAI's ChatGPT as a supportive tool in drafting and refining portions of the text.*

Course Staff

Instructors

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Teaching Assistants

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Course Coordinator

course417-admin@cs.ubc.ca - for academic concessions and scheduling questions.

Lectures Days, Time, and Place

Section	Days	Time	Location
201	Mon/Wed/Fri	10:00 AM - 11:00 AM	DMP 110
202	Mon/Wed/Fri	2:00 PM - 3:00 PM	LIFE 2201

Course Description

This course introduces students to distributed systems, computer networking, protocol design, reliability, routing, security, and cloud communication systems. Students build small networked programs and analyze how modern applications use layered communication systems.

Course Calendar

The course calendar below is intentionally formatted as a large multi-page table for parser testing. The same table continues across three pages with repeated headers and consistent grid lines.

Week #	Date	Topic	Who	Reading	In-class work	Tutorial	Quiz	Assignment
1	Jan 5	Introduction and course overview	201: Morris 202: Patel	Ch 1	ICA01 Intro systems	1. Environment setup		PA1 released
1	Jan 7	Design goals for distributed systems	201: Morris 202: Patel	1.1, 1.2	ICA02 Scalability			
1	Jan 9	Layering and abstractions	201: Morris 202: Patel	1.3	ICA03 Layers		Quiz 1	
2	Jan 12	Socket programming basics	201: Morris 202: Patel	Ch 2	ICA04 Sockets	2. TCP sockets		
2	Jan 14	Client-server applications	201: Patel 202: Patel	2.1	ICA05 Client server			
2	Jan 16	Application protocols and APIs	201: Morris 202: Patel	2.2	ICA06 APIs			PA1 due
3	Jan 19	Naming and discovery	201: Morris 202: Patel	Ch 3	ICA07 Naming	3. Protocol RFCs		PA2 released
3	Jan 21	Caching and content delivery	201: Patel 202: Patel	3.2	ICA08 Caching		Quiz 2	
3	Jan 23	Reliable message delivery	201: Morris 202: Morris	3.3	ICA09 Delivery			
4	Jan 26	Transport layer concepts	201: Morris 202: Morris	Ch 4	ICA10 Transport	4. Debugging sockets		

Week #	Date	Topic	Who	Reading	In-class work	Tutorial	Quiz	Assignment
4	Jan 28	Timeouts and retransmission	201: Patel 202: Patel	4.1	ICA11 Timeouts			PA2 due
4	Jan 30	Windowing protocols	201: Morris 202: Patel	4.2	ICA12 Windows		Quiz 3	
5	Feb 2	Congestion control	201: Morris 202: Patel	4.4	ICA13 Congestion	5. Packet traces		PA3 released
5	Feb 4	Network layer introduction	201: Morris 202: Patel	Ch 5	ICA14 Networks			
5	Feb 6	Addressing and forwarding	201: Patel 202: Patel	5.1	ICA15 Addressing			
6	Feb 9	Routing algorithms	201: Morris 202: Morris	5.2	ICA16 Routing	6. Routing lab		PA3 due
6	Feb 11	Distance vector routing	201: Morris 202: Patel	5.3	ICA17 Distance vector		Quiz 4	
6	Feb 13	Link state routing	201: Patel 202: Patel	5.4	ICA18 Link state			
7	Feb 16	Reading break - no class						
7	Feb 18	Reading break - no class						
7	Feb 20	Reading break - no class						

Week #	Date	Topic	Who	Reading	In-class work	Tutorial	Quiz	Assignment
8	Feb 23	Network security introduction	201: Morris 202: Patel	Ch 6	ICA19 Security	7. Crypto review		PA4 released
8	Feb 25	Authentication protocols	201: Morris 202: Patel	6.1	ICA20 Auth			
8	Feb 27	Encryption and TLS	201: Patel 202: Patel	6.2	ICA21 TLS		Quiz 5	
9	Mar 2	Cloud networking	201: Morris 202: Morris	Ch 7	ICA22 Cloud	8. Cloud tools		
9	Mar 4	Distributed storage systems	201: Morris 202: Patel	7.1	ICA23 Storage			PA4 due
9	Mar 6	Consensus and replication	201: Patel 202: Patel	7.2	ICA24 Consensus			
10	Mar 9	Fault tolerance	201: Morris 202: Patel	Ch 8	ICA25 Faults	9. Final review prep	Quiz 6	
10	Mar 11	Monitoring and observability	201: Morris 202: Patel	8.1	ICA26 Monitoring			PA5 released
10	Mar 13	Wrap-up and review	201: Morris 202: Patel	Review	ICA27 Review			

Grading Scheme

Assignments: 35%

Quizzes: 25%

Final exam: 30%

Participation: 10%

Textbook and References

Required textbook: Chen, R. and Wallace, P. *Distributed Systems and Network Applications*, 2nd Edition. Additional readings and tutorial resources will be posted on Canvas.

Academic Conduct

Students are expected to submit original work and follow university academic integrity policies. Collaboration rules for programming assignments will be posted with each assignment.